

the `opencv - cv2` module. The new dimensions that we want to use are 150 by 150 for height and width and three channels. A coloured image is made using three channels that are arrays of red, green and blue pixel values. Combination of these three arrays gives the pictures their different colours. We could use one channel that would read our images in grey-scale or the black and white format.

To pre-process the images you can follow these steps:

1. Create a new variable `X` that will hold the new training set. It will be an array of image pixels. Another variable `y` that will hold our training labels that are one if the image is a dog and 0 if it is a cat.
2. Next, read the images one after the other and resize them with the `cv2` commands.
3. Append 1 to `y` if the image is a dog and 0 if it is a cat.

Now to verify that all the images are correct images of dogs and cats, we will plot some of these images. Since these images are not `jpg` files anymore, therefore we cannot plot them with the `mpimg` module of `matplotlib`. image. Rather, we should use the `imshow()` command above because these are now arrays of pixels `X`.

It is time to check our labels now. Since we have a total of 4000 images (2000 cats and 2000 dogs hence our label list `y` should contain 2000 of 1s and 2000 of 0s. To plot this, follow the steps:

1. Import `seaborn` package that is another commonly used plotting package built on top of `matplotlib`. It gives very beautiful plots.

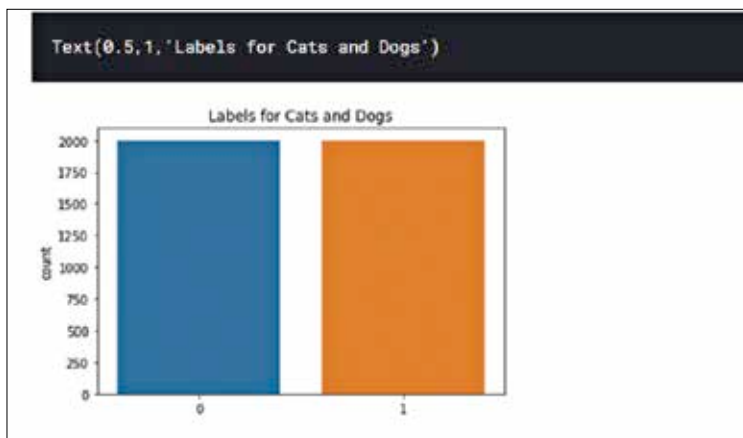


Figure: Checking for labels of data